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From Detection to Triage: Explainable Suspicious Flow Prioritization for Multiclass Intrusion Detection Using CSE-CIC-IDS2018

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Abstract

Intrusion detection systems (IDSs) are commonly evaluated through aggregate classification metrics, although operational workflows require detected flows to be interpreted, prioritized, and transformed into actionable evidence. This study proposes a detection-to-triage framework for multiclass intrusion detection using a CSE-CIC-IDS2018-derived experimental subset containing 213,463 records across one benign class and fourteen attack classes. The framework combines supervised multiclass classification, SHAP-style post hoc explanation, class-specific false positive analysis, and a Suspicious Flow Priority Score (SFPS) for analyst-oriented suspicious flow ranking. The practical role of SFPS is to reorder suspicious flows by combining model confidence, explanation strength, predefined attack severity, and validation-based false positive control, thereby producing a transparent triage list rather than a probability-only alert queue. Three detection backbones were evaluated under a shared preprocessing protocol: Random Forest, XGBoost, and a lightweight multi-layer perceptron baseline. To assess stability, experiments were repeated across five random seeds. XGBoost achieved the strongest mean performance across most aggregate indicators, with an accuracy of 0.9494 ± 0.0011 , a macro F1-score of 0.8366 ± 0.0193 , a weighted F1-score of 0.9494 ± 0.0011 , and a Matthews Correlation Coefficient of 0.9429 ± 0.0012 . Random Forest produced closely comparable results, while the lightweight MLP remained lower on aggregate and macro-level indicators. False positive analysis showed that the alert burden was concentrated in selected classes and differed across models, confirming that aggregate performance alone is insufficient for assessing IDS usefulness. SHAP-style analysis identified stable flow-level contributors to XGBoost discrimination, while SFPS substantially changed the post-detection ordering of suspicious flows compared with probability-only ranking. The study does not claim universal state-of-the-art superiority, causal explanation, or deployment validation; instead, it demonstrates how multiclass IDS outputs can be extended into explainable, false positive-aware, and triage-oriented rankings for analyst review.



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Keywords: alert prioritization; CSE-CIC-IDS2018; explainable AI; intrusion detection system; machine learning; suspicious flow ranking

1. Introduction

Network intrusion detection systems (IDSs) remain a central component of contemporary cybersecurity architectures, particularly as digital infrastructures are exposed to increasingly diverse, high-volume, and adaptive network threats. Over the past two decades, intrusion detection research has evolved from rule-based and statistical anomaly

detection toward machine learning and deep learning approaches capable of modeling nonlinear, high-dimensional, and heterogeneous traffic patterns [1,2]. This development has been accompanied by systematic efforts to compare anomaly-based network intrusion detection methods, benchmark datasets, feature representations, and evaluation protocols across different attack categories and network environments [3]. More recently, deep learning-driven IDS research has further expanded the methodological landscape by introducing architectures designed to learn complex representations from large-scale network traffic data [4].

Despite this progress, a persistent limitation remains in the way IDS performance is commonly framed. Many studies continue to treat detection as the system's final objective, emphasizing aggregate classification metrics such as accuracy, precision, recall, or F1-score. Although these metrics are necessary for evaluating statistical performance, they do not fully capture the operational reality of intrusion detection. In practical security environments, IDS outputs must be interpreted, prioritized, and transformed into actionable evidence. A model that correctly assigns a traffic flow to an attack class is therefore only partially useful if it does not help analysts understand why the flow was classified as suspicious, how urgent it is, and which alerts to investigate first.

This issue becomes especially important in light of recent technically advanced IDS architectures. For instance, Liu et al. proposed DLG-IDS, a dynamic graph and LLM-semantic enhanced spatio-temporal graph neural network for lightweight intrusion detection in industrial control systems [5]. Their framework integrates dynamic graph construction, semantic enhancement, sparse graph attention, and local-window transformer mechanisms to address the specific requirements of industrial control environments. Such work illustrates the increasing sophistication of modern IDS research and the growing emphasis on detection accuracy, false positive reduction, computational efficiency, and operational feasibility. However, not every IDS problem requires industrial-control-specific graph modeling, large language model integration, or highly complex spatio-temporal architectures. A complementary research direction is needed: one that focuses on transforming multiclass IDS outputs into explainable and prioritized evidence for analyst-oriented triage.

This operational perspective is crucial because automation alone does not guarantee better security decision-making. Prior research has shown that excessive or poorly calibrated automation can negatively affect user performance in simulated intrusion detection tasks [6]. Earlier work on the role of security analysts similarly emphasized that intrusion detection is not merely a technical classification process, but a form of analytical work involving interpretation, contextual reasoning, prioritization, and decision-making under uncertainty [7]. In this sense, the practical value of an IDS model depends not only on whether it detects malicious traffic, but also on whether its outputs can support human analysts who must manage large volumes of uncertain and potentially redundant alerts.

The problem is intensified by alert fatigue in security operations centers. When analysts are exposed to excessive alert volumes, many of which may be false positives, low-priority events, or insufficiently contextualized detections, their ability to identify truly critical threats can deteriorate [8]. Recent research on alert prioritization has therefore emphasized the need for methods that rank, contextualize, and reduce the cognitive burden of alert handling [9]. This aligns with contemporary incident response and cybersecurity risk management guidance, which stresses prioritization, contextual assessment, and the efficient allocation of response resources as essential elements of operational cybersecurity practice [10,11].

Against this background, this study proposes an explainable suspicious flow prioritization framework for multiclass intrusion detection using the CSE-CIC-IDS2018 benchmark dataset. Rather than introducing another high-complexity IDS architecture, the study ad-

dresses a narrower but operationally significant problem: how to move from multiclass detection to explainable triage. The proposed framework combines supervised multiclass classification, SHAP-based post hoc explainability, and a Suspicious Flow Priority Score designed to rank detected flows according to model confidence, explanation strength, attack severity, and false positive control considerations. In this way, intrusion detection is treated not as the endpoint of the analytical process, but as the first stage in constructing ranked, interpretable, and analyst-oriented evidence.

The main objective of this study is to develop and evaluate a reproducible public dataset framework that connects multiclass IDS classification with explainable suspicious flow prioritization. The study does not aim to claim universal state-of-the-art superiority, real-time deployment validation, or full forensic reconstruction. Instead, it focuses on a scope-controlled contribution: improving the interpretability and operational usefulness of multiclass IDS outputs by linking detection results to transparent feature-level explanations and triage-oriented ranking.

Accordingly, the study is positioned not as a general state-of-the-art leaderboard, but as a scope-controlled detection-to-triage framework. Its purpose is to evaluate how multiclass IDS outputs can be transformed into explainable, false positive-aware suspicious flow rankings that support analyst-oriented triage. Within this framework, SFPS is introduced as a transparent post-detection ranking mechanism that combines classification confidence, SHAP-based explanation strength, predefined attack severity, and validation-based false positive control. Its practical value lies in changing a probability-only alert queue into a structured triage list that makes the ranking logic explicit for analyst review.

The study is guided by the following research questions:

RQ1. How effectively can selected machine learning and lightweight deep learning models perform multiclass intrusion detection on the CSE-CIC-IDS2018 dataset?

RQ2. Which network-flow features contribute most strongly to attack-type attribution, and how can these contributions be explained using SHAP-based post hoc explainability?

RQ3. How does a Suspicious Flow Priority Score change the ordering of suspicious flows compared with classification probability alone?

RQ4. How does false positive burden affect the operational usefulness of multiclass IDS outputs?

The contributions of this paper are fourfold. First, it presents a reproducible multiclass IDS pipeline on the public CSE-CIC-IDS2018 benchmark dataset. Second, it integrates SHAP-based explainability to support both global and local interpretation of attack-type attribution. Third, it proposes a Suspicious Flow Priority Score that operationalizes suspicious flow ranking by explicitly combining model confidence, SHAP-based explanation strength, predefined attack severity, and validation-based false positive control, thereby distinguishing triage priority from classification probability alone. Fourth, it evaluates IDS outputs not only through conventional classification metrics, but also through false positive burden analysis and evidence prioritization, thereby addressing the gap between statistical detection performance and operational alert usability.

The remainder of the paper is organized as follows. Section 2 reviews related work on IDS benchmark datasets, CSE-CIC-IDS2018-based intrusion detection, explainable AI in cybersecurity, and alert prioritization. Section 3 describes the dataset, preprocessing procedure, classification models, explainability strategy, Suspicious Flow Priority Score, and evaluation metrics. Section 4 presents the experimental results. Section 5 discusses the operational implications, limitations, and positioning of the proposed framework. Section 6 concludes the paper.

2. Related Work

2.1. Benchmark IDS Datasets and the Role of CSE-CIC-IDS2018

The development of reliable intrusion detection systems depends strongly on the availability of representative benchmark datasets. Early IDS datasets have been widely criticized for limited attack diversity, outdated traffic profiles, insufficient realism, and weak correspondence with contemporary network environments. As a result, the construction of more realistic datasets has become a central methodological concern in IDS research. Sharafaldin et al. addressed this issue by proposing a modern intrusion detection dataset generation approach focused on realistic traffic characterization and diverse attack scenarios [12]. The CSE-CIC-IDS2018 dataset, released by the Canadian Institute for Cybersecurity, extends this line of work by providing large-scale network traffic data with multiple attack categories, flow-level features, and a public benchmark structure suitable for machine learning-based evaluation [13].

The importance of such datasets is reinforced by earlier work on systematic benchmark generation. Shiravi et al. argued that IDS datasets should be built through a structured process that captures normal and malicious behavior in a controlled yet realistic environment [14]. This dataset-centric perspective is also reflected in the development of UNSW-NB15, which was introduced as a comprehensive network intrusion detection dataset containing modern attack types and synthetic-normal traffic interactions [15]. Although UNSW-NB15 remains an important benchmark, the present study focuses on CSE-CIC-IDS2018 because of its scale, flow-based structure, attack diversity, and extensive use in recent IDS research.

Several reviews have emphasized that dataset choice significantly shapes reported IDS performance. Thakkar and Lohiya reviewed the advancement of intrusion detection datasets and highlighted the need for benchmark resources that reflect evolving network conditions and attack strategies [16]. Ring et al. similarly showed that network-based IDS datasets differ substantially in terms of traffic generation, feature availability, attack representation, and suitability for model comparison [17]. Hindy et al. further demonstrated that dataset design affects not only detection accuracy but also the taxonomy of threats that models are able to learn and distinguish [18]. These findings are important for the present study because they suggest that IDS evaluation cannot be interpreted independently of dataset structure, feature space, class distribution, and attack labeling.

Within this context, CSE-CIC-IDS2018 has become one of the most frequently used datasets for large-scale IDS evaluation. Leevy et al. provided a survey and analysis of intrusion detection models based on CSE-CIC-IDS2018, emphasizing both its value as a large public benchmark and the methodological challenges associated with class imbalance, feature dimensionality, and model comparability [19]. This makes CSE-CIC-IDS2018 particularly suitable for the present study, which does not aim merely to maximize classification performance, but to examine how multiclass detection outputs can be made explainable and operationally useful for suspicious flow prioritization.

2.2. Multiclass IDS Models on CSE-CIC-IDS2018

Deep learning has become a dominant direction in recent IDS research because of its capacity to model nonlinear relations, high-dimensional feature spaces, and complex attack patterns. Ferrag et al. provided an extensive comparative analysis of deep learning approaches, datasets, and cybersecurity intrusion detection methods, showing that neural architectures can achieve strong benchmark performance but often differ substantially in terms of generalizability, computational cost, and evaluation consistency [20]. This observation is important because the present study does not treat deep learning complexity

as an end in itself. Instead, it uses selected models as part of a broader detection-to-triage pipeline.

A growing number of studies have evaluated CSE-CIC-IDS2018 using deep learning or hybrid optimization strategies. Alzughaibi and El Khediri proposed a cloud IDS based on deep neural networks combined with particle swarm optimization on CSE-CIC-IDS2018, illustrating the tendency to improve detection through optimized neural configurations [21]. Farhan and Jasim analyzed the performance of deep learning models on the same dataset, further confirming its role as a benchmark for evaluating IDS architectures under complex attack distributions [22]. Songma et al. introduced a three-phase optimization approach for CSE-CIC-IDS2018, showing that preprocessing, feature selection, and model optimization can materially affect detection outcomes [23].

Feature relevance is another recurring issue in CSE-CIC-IDS2018-based research. Göcs and Johanyák examined relevant features of the dataset for IDS development, emphasizing that performance does not depend solely on classifier selection, but also on the quality and interpretability of the selected flow attributes [24]. This is directly relevant to the present study because the proposed framework relies on explainability not only to justify predictions, but also to identify which network-flow features contribute most strongly to attack-type attribution. Hewapathirana further examined two-stage intrusion detection using modern machine learning approaches on CSE-CIC-IDS2018, reinforcing the value of structured pipelines rather than single-step classification alone [25].

Recent studies also show the continued relevance of conventional machine learning classifiers in IDS evaluation. Ibrahim et al. investigated machine learning classifiers on CSE-CIC-IDS2018, demonstrating that classical models can remain competitive when combined with appropriate preprocessing and evaluation protocols [26]. Coşar et al. similarly evaluated machine learning methods on CSE-CIC-IDS2018, contributing to the growing evidence that interpretable and computationally efficient models remain useful in benchmark IDS settings [27]. These studies support the methodological choice of including tree-based machine learning models in the present work rather than relying exclusively on deep learning architectures.

At the same time, recent deep learning studies continue to expand the CSE-CIC-IDS2018 literature. Boudaine et al. proposed a deep learning-based anomaly and intrusion detection approach using this dataset [28], while Alzubi et al. combined advanced feature selection and machine learning techniques to optimize IDS performance on CSE-CIC-IDS2018 [29]. Collectively, these works demonstrate that the dataset has already been widely used for detection-focused evaluation. However, most existing studies still concentrate primarily on classification performance. The present study therefore differs by focusing on what happens after multiclass classification: whether detected flows can be ranked, explained, and transformed into analyst-oriented evidence.

2.3. Explainable AI for Intrusion Detection

As IDS models become more complex, explainability has become increasingly important for understanding, validating, and operationalizing model outputs. Lundberg and Lee introduced SHAP as a unified framework for interpreting model predictions by assigning additive feature contributions to individual predictions [30]. Ribeiro et al. earlier proposed LIME as a model-agnostic explanation method designed to answer why a classifier produced a specific prediction [31]. These approaches are especially relevant in cybersecurity, where analysts often need to understand not only what a model predicted, but which traffic characteristics drove that prediction.

The broader XAI literature provides the conceptual foundation for this shift. Barredo Arrieta et al. systematized the main concepts, taxonomies, opportunities, and challenges

of explainable artificial intelligence, emphasizing that explainability is essential when automated decisions affect trust, accountability, and human oversight [32]. In intrusion detection, this is particularly important because black-box predictions may be difficult to verify, prioritize, or translate into response actions. Patil et al. specifically examined explainable AI for IDS, showing that XAI methods can improve transparency and support the interpretation of security alerts [33]. Capuano et al. extended this perspective by surveying explainable AI in cybersecurity more broadly, demonstrating that explainability is becoming a key requirement for security analytics, threat detection, and decision support [34].

Recent cybersecurity-focused surveys further reinforce this point. Rjoub et al. reviewed explainable AI for cybersecurity and highlighted its relevance for improving the trust in, and transparency and operational adoption of intelligent security systems [35]. Mohale and Obagbuwa systematically reviewed the integration of XAI into IDS, emphasizing that interpretability is especially important when detection models are deployed in environments where false alarms, missed detections, and analyst trust directly affect security outcomes [36]. Samed and Sagiroglu similarly discussed explainable AI models in IDS, positioning XAI as a methodological requirement rather than a merely optional visualization layer [37].

Several recent studies apply XAI directly to intrusion detection contexts. Hosain and Çakmak proposed an XAI-XGBoost approach for securing Internet of Medical Things systems, showing how explainable tree-based models can support cybersecurity interpretation in sensitive environments [38]. Farooqui et al. introduced AutoSHARC, a SHAP-guided explainable IDS framework using post hoc retraining, demonstrating how SHAP explanations can be incorporated into IDS refinement and feedback processes [39]. Nezhadsistani et al. proposed a robust and explainable IDS framework for 5G/B5G networks, further indicating that explainability is becoming a central design criterion in modern IDS architectures [40].

These studies establish the relevance of XAI for intrusion detection, but they also reveal an unresolved gap. In many IDS studies, explainability remains confined to global feature importance plots or post hoc visual inspection. The present study extends this logic by using SHAP not only to explain model behavior, but also to support suspicious flow prioritization. In other words, explanations are treated as part of the evidence structure used to rank alerts, not simply as descriptive outputs added after classification.

2.4. Alert Prioritization and the Detection-to-Triage Gap

Although classification performance remains important, operational cybersecurity increasingly requires systems that can reduce alert overload and support analyst decision-making. Wang et al. proposed AlertPro, a context-aware alert prioritization approach based on reinforcement learning for multi-step attack detection, showing that alert handling benefits from ranking mechanisms that account for attack context and operational relevance [41]. This direction is closely aligned with the present study, which treats IDS outputs as inputs to a prioritization process rather than as final decisions.

Alert correlation is another important component of operational threat detection and response. Lanigan et al. reviewed alert correlation methods for intelligent threat detection and response, emphasizing the need to connect isolated detection outputs into more meaningful and actionable structures [42]. Although the present study does not implement full alert correlation or multi-stage attack reconstruction, it follows the same operational logic by ranking suspicious flows according to multiple evidence dimensions. This makes the proposed framework narrower than a complete SOC automation system, but more operationally meaningful than classification alone.

Alert fatigue remains one of the most significant obstacles to effective security monitoring. Voutsas et al. examined machine learning approaches for mitigating alert fatigue in cloud monitoring systems, highlighting how prioritization and filtering can reduce unnecessary analyst burden [43]. Yu et al. provided a broader survey on the intelligent management of alerts and incidents in IT services, showing that modern alert handling requires ranking, contextualization, incident grouping, and intelligent escalation mechanisms [44]. These works support the idea that IDS evaluation should include not only detection accuracy but also the operational usability of alerts.

Finally, Giarimpampa et al. reviewed the role of artificial intelligence in enhancing security operations, showing that AI-based security tools must be assessed not only by their predictive performance but also by their ability to support operational workflows, analyst reasoning, and incident response processes [45]. This perspective frames the central motivation of the present study. The goal is not to replace analysts or claim full SOC validation, but to provide a reproducible framework that connects multiclass IDS predictions with explainable suspicious flow ranking. In this sense, the proposed approach addresses the detection-to-triage gap: the underexplored space between model output and operational decision support.

To clarify the positioning of the proposed framework, Table 1 compares representative strands of related work with the present study. Existing SHAP- and XAI-based IDS studies primarily strengthen interpretability, while alert prioritization studies focus on ranking, correlation, or operational alert handling. The present framework connects these strands at the flow level by combining multiclass IDS outputs, SHAP-style explanation, validation-based false positive control, and predefined attack severity weighting into a single suspicious flow ranking procedure. This comparison highlights that the contribution is not a new detection backbone alone, but a post-detection triage layer that makes the ranking logic explicit and reproducible.

Table 1. Positioning the proposed framework relative to related IDS, XAI, and alert prioritization studies.

Study/Strand	Main Focus	XAI/SHAP Component	Alert Ranking or Triage Component	FP/Severity Awareness	Difference from the Present Study
Hosain and Çakmak [38]	Explainable XGBoost-based detection	Yes	No explicit triage ranking	Limited	Explainability is not integrated into a ranking score.
Farooqui et al. [39]	SHAP-guided IDS refinement	Yes	Indirect	Model refinement focus	SHAP supports retraining, not post-detection triage.
Nezhadsistani et al. [40]	Robust explainable IDS	Yes	Not central	Limited	No unified severity- and FPR-aware flow ranking.
Wang et al. [41]	Context-aware alert prioritization	Not SHAP-focused	Yes	Context-aware	Broader alert prioritization; not flow-level SHAP-SFPS ranking.
Lanigan et al. [42]	Alert correlation and response	Not SHAP-focused	Yes	Operational	Focuses on correlation, not multiclass flow-level scoring.
Voutsas et al. [43]; Yu et al. [44]	Alert fatigue and incident management	Not SHAP-focused	Yes	Alert burden focus	Addresses alert management, not IDS-derived SFPS scoring.
Giarimpampa et al. [45]	AI support for security operations	Broad AI/SOC	Decision-support focus	Workflow level	Provides SOC context, not a reproducible flow ranking formula.
Present study	Detection-to-triage framework	Yes	Yes, SFPS ranking	Yes	Integrates classification, SHAP, severity, and validation FPR.

2.5. Summary of the Research Gap

The reviewed literature shows that IDS research has made substantial progress in dataset construction, CSE-CIC-IDS2018-based benchmarking, machine and deep learning classification, explainable AI, and alert prioritization. These strands often remain sepa-

rated. Benchmark studies usually focus on classification metrics; XAI studies often stop at explanation; and alert prioritization studies are commonly situated in broader SOC or incident-management contexts rather than directly linked to flow-level multiclass IDS outputs. As a result, there remains a methodological gap between detecting an attack type, explaining the prediction, and ranking the suspicious flow for analyst-oriented triage.

The present study addresses this gap by integrating these components into a single scope-controlled framework. It uses CSE-CIC-IDS2018 as a public benchmark dataset, evaluates multiclass IDS models, applies SHAP-based explainability, and proposes a Suspicious Flow Priority Score that combines model confidence, explanation strength, attack severity, and validation-based false positive control adjustment. As summarized in Table 1, the contribution lies in connecting explainable IDS interpretation with suspicious flow ranking, rather than treating classification, explanation, and alert prioritization as separate tasks. This positioning allows the study to contribute beyond conventional accuracy-driven IDS evaluation while avoiding unsupported claims of real-time deployment, full forensic reconstruction, or universal state-of-the-art superiority.

3. Materials and Methods

3.1. Experimental Design and Dataset Preparation

This study adopts a controlled benchmark design to evaluate how multiclass intrusion detection outputs can be transformed into explainable suspicious flow rankings. The experimental pipeline consists of four main stages: dataset preparation, multiclass model training, SHAP-based explanation, and suspicious flow prioritization.

The complete detection-to-triage workflow is summarized in Figure 1.

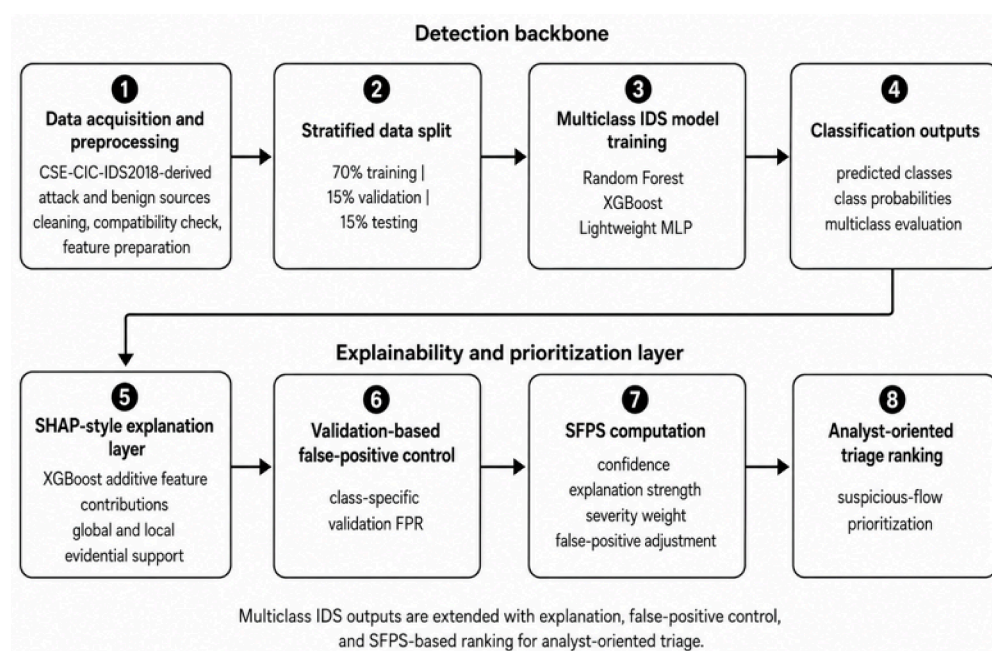


Figure 1. Detection-to-triage experimental pipeline.

The workflow starts with CSE-CIC-IDS2018-derived data acquisition and preprocessing, followed by stratified train/validation/test splitting and multiclass model training using Random Forest, XGBoost, and a lightweight MLP baseline. Classification outputs are then extended through an XGBoost SHAP-style explanation layer and validation-based false positive control. These components are combined in the Suspicious Flow Priority Score (SFPS), which supports analyst-oriented triage ranking of suspicious flows. The aim is not to introduce another high-complexity IDS architecture, but to examine whether

standard and computationally tractable models can be extended with an explanation- and triage-oriented layer that improves the operational interpretability of IDS outputs.

The empirical dataset was constructed from two compatible CSE-CIC-IDS2018-derived sources. Attack-class records were obtained from the cleaned CSV release, while benign flow records were extracted from the official processed CSE-CIC-IDS2018 traffic file [12,13]. Column compatibility was verified before modeling; the attack-class files contained 79 columns, while the official processed file contained 80 columns. The only additional field in the official processed file was the timestamp field, which was excluded from modeling. Both sources contained the same label field and the same flow-based feature structure after timestamp removal. The resulting dataset therefore enables multiclass evaluation across benign and attack categories. Unlike binary IDS formulations that collapse all attacks into a single malicious class, this study preserves the multiclass structure of the dataset.

Let $X = \{x_1, x_2, \dots, x_n\}$ denote the set of preprocessed network-flow records, where each x_i is represented by a vector of flow-level features. Let $Y = \{y_1, y_2, \dots, y_n\}$ denote the corresponding class labels, where $y_i \in \{1, \dots, K\}$, and K represents the number of retained traffic classes after preprocessing. The multiclass classification task is defined as learning a mapping:

$$f: X \rightarrow Y \quad (1)$$

where the model outputs a class probability distribution:

$$P(y = k \mid x_i), \quad k = 1, \dots, K \quad (2)$$

The predicted class is then defined as

$$\hat{y}_i = \underset{k}{\operatorname{argmax}} P(y = k \mid x_i) \quad (3)$$

Only flows predicted as non-benign are passed to the prioritization stage. The prioritization layer is therefore not a replacement for classification, but a post-detection mechanism that ranks suspicious flows for analyst-oriented review.

Preprocessing was designed to produce a clean, leakage-controlled, and model-consistent feature representation. Non-informative identifiers, duplicate records, missing values, infinite values, and fields that could introduce direct label leakage were removed before model training. Numerical variables were standardized using z-score normalization. Normalization parameters were estimated exclusively on the training split and then applied unchanged to the validation and test sets. This procedure reduces information leakage and ensures that the test set remains unseen during preprocessing.

Categorical variables, where present in the retained dataset version, were transformed using one-hot encoding. The same transformed feature matrix was used for all models. This shared-input design allows model differences to be interpreted as differences in model behavior rather than artifacts of inconsistent preprocessing. The dataset was divided into training, validation, and test subsets using stratified sampling to preserve the class distribution across all splits:

70% training, 15% validation, 15% testing.

Stratification is particularly important in multiclass IDS evaluation because rare attack categories may otherwise be underrepresented in the validation or test subset. The validation set was used for model selection and estimation of false positive control parameters. The test set was reserved for final classification evaluation, SHAP explanation examples, and suspicious flow prioritization results.

The retained class structure after preprocessing is reported in Table 2. This table is necessary because CSE-CIC-IDS2018 is highly class-imbalanced, and class distribution directly affects the interpretation of multiclass IDS performance, false positive behavior, and prioritization results. After removal of missing and infinite values, the final retained experimental subset contained 213,463 records and 79 columns, covering one benign class and fourteen attack classes.

Table 2. Dataset composition after preprocessing.

Class	Retained Records
DDoS attack-HOIC	20,000
FTP-BruteForce	20,000
DoS attacks-SlowHTTPTest	20,000
Infiltration	20,000
DoS attacks-Hulk	20,000
DoS attacks-GoldenEye	20,000
Bot	20,000
DDoS attacks-LOIC-HTTP	20,000
SSH-Bruteforce	20,000
Benign	19,815
DoS attacks-Slowloris	10,990
DDoS attack-LOIC-UDP	1730
Brute Force-Web	611
Brute Force-XSS	230
SQL Injection	87
Total	213,463

The capped sampling strategy was used to preserve multiclass attack diversity while preventing high-frequency attack categories from dominating model training and aggregate evaluation. For attack classes with more than 20,000 available records, a maximum of 20,000 records was retained. Classes with fewer available records were retained in full, including Brute Force -Web, Brute Force -XSS, and SQL Injection. This decision was made to avoid further reducing already rare attack categories and to allow their minority-class behavior to remain visible in per-class evaluation. Consequently, the retained subset should be interpreted as a controlled multiclass experimental subset rather than a frequency-preserving reproduction of the full CSE-CIC-IDS2018 distribution. Because SQL Injection contains only 87 retained records, the results for this class are reported and interpreted explicitly as minority-class evidence rather than as a basis for broad operational generalization.

To complement the tabular overview, Figure 2 visualizes the retained class distribution after preprocessing and capped stratified sampling. The logarithmic scale is used because the retained subset contains both capped high-frequency classes and rare attack classes retained in substantially smaller numbers. This visualization is important for interpreting the subsequent model evaluation, since class imbalance may affect macro-averaged performance, minority-class behavior, and class-specific false positive burden.

The figure shows the final retained class structure of the CSE-CIC-IDS2018-derived experimental subset after preprocessing and capped stratified sampling. A logarithmic x-axis is used to preserve visibility of both capped high-frequency classes and rare attack classes.

The resulting class structure provides the empirical basis for the classification, explainability, and prioritization stages described in the following subsection. Because the dataset preserves one benign class and fourteen attack classes, the experimental design supports multiclass evaluation rather than a simplified binary benign-versus-attack formulation.

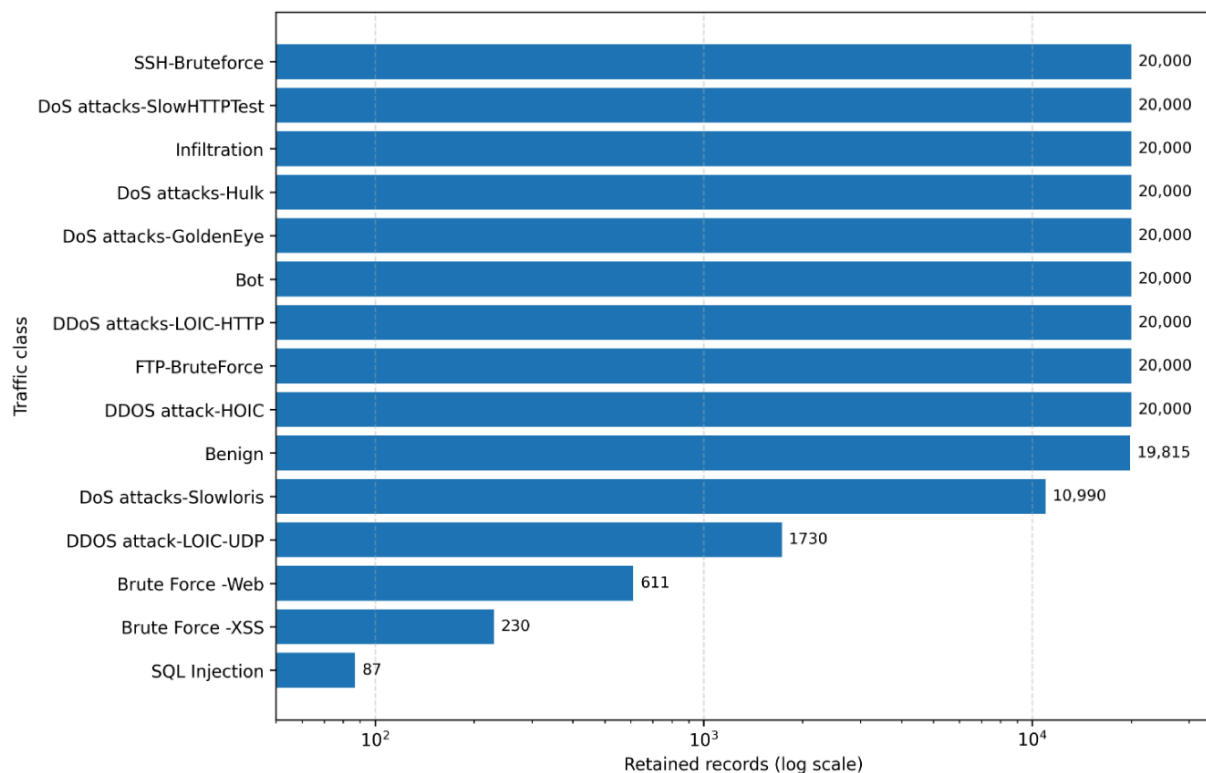


Figure 2. Retained class distribution after preprocessing and capped stratified sampling.

3.2. Classification, Explainability, and Suspicious Flow Prioritization Framework

Three model families are evaluated: Random Forest, XGBoost, and a lightweight deep learning baseline. These models were selected to balance predictive performance, interpretability, computational feasibility, and methodological relevance to structured flow-based IDS data. Random Forest is included as a robust tree-based ensemble method capable of modeling nonlinear feature interactions while remaining comparatively interpretable [46]. Gradient boosting provides the conceptual basis for sequential ensemble learning [47], while XGBoost is included as a scalable and regularized implementation of gradient-boosted trees that is widely used for structured-data classification [48].

A lightweight deep learning baseline is included to provide a compact neural comparator. Its purpose is not to introduce a new neural architecture or compete with transformer-, graph-, or LLM-based IDSs. Instead, it provides a controlled baseline for evaluating whether a lightweight neural model behaves differently from tree-based models under the same preprocessing and evaluation conditions. Recent IDS studies show that recurrent, optimized, and lightweight deep learning models can achieve strong performance in multi-class or complex traffic settings [49–52]. In this study, however, the neural model is treated as a baseline within a classification-and-triage framework, not as the main contribution.

Because the lightweight MLP was included as a compact neural baseline, it was trained under the same capped and stratified data protocol as the tree-based models, but without an additional class-weighting mechanism. This design keeps the MLP role limited to a controlled comparator rather than an optimized imbalance-specific neural architecture. Its macro-level performance is therefore interpreted with particular attention to minority-class behavior, especially for rare categories such as SQL Injection.

All models produce class probability estimates. These estimates are used in two ways: first, to evaluate multiclass detection performance; second, to compute the model confidence component of the Suspicious Flow Priority Score. The classification models are therefore used as controlled detection backbones within the proposed framework. The

purpose of their comparison is not to establish universal model superiority, but to provide a stable predictive basis for evaluating the added value of SHAP-based explanation and SFPS-based prioritization. Random Forest, XGBoost, and the lightweight deep learning baseline are therefore treated as comparable detection engines within the same triage-oriented pipeline, not as competing state-of-the-art claims.

To interpret model predictions, the study applies SHAP-based post hoc explainability. SHAP values estimate the contribution of each feature to a given prediction by assigning additive feature-attribution scores [30]. In this study, SHAP is used at two levels. First, global SHAP analysis identifies the most influential features across the dataset and across attack classes. Second, local SHAP analysis explains individual suspicious flows by identifying the top features that contributed most strongly to the predicted attack class.

For each suspicious flow, the explanation is represented as a vector of feature-level contributions associated with the predicted class:

$$\Phi_i = \{\phi_{i1}, \phi_{i2}, \dots, \phi_{im}\} \quad (4)$$

where ϕ_{ij} denotes the SHAP contribution of feature j for prediction i , and m denotes the number of retained input features. Positive SHAP values increase the model output toward the predicted class, whereas negative values reduce it. Because the prioritization procedure requires a scalar measure of explanation strength, the absolute SHAP contributions of the top m most influential features are aggregated:

$$E_i^{raw} = \sum_{j \in T_m} |\phi_{ij}| \quad (5)$$

where T_m denotes the subset of the top m features with the largest absolute SHAP contributions for flow x_i . The use of absolute values reflects the magnitude of explanatory support, regardless of whether an individual feature increases or decreases the probability of the predicted class.

Since raw SHAP magnitudes are model- and dataset-dependent, the resulting explanation strength score is normalized to the unit interval:

$$E_i = \frac{E_i^{raw} - E_{min}}{E_{max} - E_{min} + \epsilon} \quad (6)$$

where E_{min} and E_{max} are estimated on the validation set, and ϵ is a small constant used to avoid division by zero. This validation-based normalization prevents the score from being calibrated directly on the test set. The resulting value E_i is interpreted as a relative explanation strength indicator, not as an absolute measure of forensic certainty.

The central methodological contribution of this study is the Suspicious Flow Priority Score (SFPS). The SFPS is designed to rank suspicious flows by combining four complementary components: model confidence, SHAP-based explanation strength, predefined attack severity, and validation-based false positive control. The score is not intended to represent absolute incident severity, forensic certainty, or to replace analyst judgment. Instead, it functions as a reproducible triage-oriented ranking heuristic that converts multiclass IDS outputs into an ordered evidence structure.

For each suspicious flow x_i , the model confidence component is defined as

$$C_i = \max_k P(y = k | x_i) \quad (7)$$

where C_i denotes the highest predicted class probability assigned by the classifier. This component captures the model's internal confidence in the predicted class, but it is not

sufficient on its own for operational prioritization because high confidence may still occur for classes associated with frequent false positives.

The attack severity component is defined as a predefined class-level weight:

$$S_i \in [0, 1] \quad (8)$$

where S_i represents the severity weight assigned to the predicted attack class. The severity weights were predefined before test-set evaluation and used only as class-level triage weights in the SFPS computation. They do not represent ground-truth incident impact, were not learned from the test data, and should be interpreted only as operational prioritization parameters.

The class-level severity mapping used in the SFPS computation is reported in Table 3. The weights were assigned as predefined operational triage parameters on a 0–1 scale, where higher values indicate greater assumed priority for analyst review. The mapping reflects the expected operational consequences of each attack category, including potential service disruption, credential compromise, application-layer exploitation, and unauthorized access. Benign traffic was assigned a weight of zero because it is not a suspicious flow priority target. These weights are not presented as universal cyber risk values; they are transparent scoring parameters used to make the SFPS computation reproducible.

Table 3. Predefined class-level severity weights used in the SFPS computation.

Class	Severity Weight	Rationale
Benign	0.00	Normal traffic; not a suspicious flow priority target.
Bot	0.70	May indicate compromised hosts and automated command behavior.
Brute Force-Web	0.70	Credential/application attack relevant for access compromise.
Brute Force-XSS	0.70	Application-layer exploit attempt; operationally relevant despite low frequency.
DDoS attack-HOIC	0.90	High-volume denial-of-service activity with potentially severe service impact.
DDoS attack-LOIC-UDP	0.85	UDP-based denial-of-service activity with service disruption potential.
DDoS attacks-LOIC-HTTP	0.90	HTTP-based distributed denial-of-service activity with high service impact.
DoS attacks-GoldenEye	0.80	Denial-of-service attack with potential availability impact.
DoS attacks-Hulk	0.95	High-impact denial-of-service behavior prioritized for availability risk.
DoS attacks-SlowHTTPTest	0.75	Slow-rate DoS behavior; operationally relevant but context-dependent.
DoS attacks-Slowloris	0.75	Slow-rate DoS behavior; operationally relevant but context-dependent.
FTP-BruteForce	0.70	Credential attack against FTP services; relevant for access compromise.
Infiltration	1.00	Potential unauthorized access or lateral movement indicator; highest operational concern.
SQL Injection	0.90	Application-layer exploit attempt with potential data compromise consequences.
SSH-Bruteforce	0.70	Credential attack against remote-access services; important but commonly noisy.

The sensitivity analysis later examines whether alternative weighting configurations change the ranking pattern, while the main SFPS results use the predefined weights in Table 3 together with equal component weights.

The false positive control adjustment is defined as

$$A_i = 1 - FPR_{y_i}^{val} \quad (9)$$

where FPR denotes the validation-set false positive rate for the predicted class, calculated as

$$FPR_{\hat{y}_i}^{val} = \frac{FP_{\hat{y}_i}^{val}}{FP_{\hat{y}_i}^{val} + TN_{\hat{y}_i}^{val}} \quad (10)$$

This adjustment reduces the priority of classes that the model tends to over-predict incorrectly on the validation set. In this way, the prioritization score incorporates not only prediction confidence but also class-specific reliability.

$$SFPS_i = w_c C_i + w_e E_i + w_s S_i + w_f A_i \quad (11)$$

where w_c , w_e , w_s , and w_f denote the weights assigned to model confidence, SHAP evidence strength, attack severity, and false positive adjustment, respectively. The weights satisfy $w_c + w_e + w_s + w_f = 1$, with $w_c, w_e, w_s, w_f \geq 0$. In the main analysis, an equal-weight configuration was used, with $w_c = w_e = w_s = w_f = 0.25$, to avoid tailoring the prioritization score to a specific operational scenario. Alternative weighting schemes were examined in the sensitivity analysis, while the primary reported SFPS results are based on the equal-weight configuration.

The SFPS should therefore be interpreted as a ranking mechanism rather than as an absolute measure of cyber risk. A flow with high classification probability but weak explanation support or poor validation-based false positive reliability may receive a lower priority than a flow with slightly lower probability but stronger explanation evidence, higher predefined severity, and better class-specific reliability. This distinction is central to the proposed framework: classification confidence and triage priority are related but not identical.

3.3. Evaluation Metrics, False Positive Burden, and Implementation Details

The evaluation strategy is designed to assess both multiclass detection performance and triage-oriented usability. Conventional classification metrics are necessary for measuring predictive quality, but they are insufficient on their own because an IDS model may achieve strong aggregate performance while still producing an operationally problematic number of false positives. Therefore, this study evaluates model behavior at two levels: first, as a multiclass classifier, and second, as a generator of suspicious flow outputs that must be ranked for analyst review. Accordingly, the evaluation is structured around two complementary comparisons. The first comparison assesses model-level detection performance across Random Forest, XGBoost, and the lightweight deep learning baseline. The second comparison assesses prioritization behavior by comparing probability-only ranking with the proposed SFPS-based ranking. This two-layer evaluation design ensures that the study does not stop at classification accuracy, but directly tests whether explanation strength and validation-based false positive reliability can change the operational ordering of suspicious flows.

Because CSE-CIC-IDS2018 contains uneven class distributions, accuracy is not interpreted as a standalone indicator of performance. The study reports accuracy, precision, recall, F1-score, macro-F1, weighted-F1, Matthews Correlation Coefficient (MCC), ROC-AUC, PR-AUC, and class-specific false positive rates, where applicable. Macro-averaged metrics assign equal importance to all classes, whereas weighted metrics account for class frequency. Class-specific false positive rates are calculated in a one-vs-rest manner.

Precision measures the proportion of correctly identified attack instances among all instances predicted as attacks:

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

where TP denotes true positives and FP denotes false positives. In IDS evaluation, precision is operationally important because excessive false positives increase alert volume and may reduce analyst trust in automated detections.

Recall measures the proportion of correctly detected attack instances among all actual attack instances:

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

where FN denotes false negatives. High recall is important because missed attacks may remain undetected and therefore unresolved.

The F1-score is the harmonic mean of precision and recall:

$$F1 = \frac{2PR}{P + R} \quad (14)$$

where P denotes precision, and R denotes recall.

This metric is useful when precision and recall must be interpreted jointly, especially under class imbalance.

Accuracy measures the proportion of correctly classified instances among all evaluated instances:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

where TN denotes true negatives. Although accuracy is commonly reported, it may be misleading in imbalanced IDS datasets and is therefore interpreted together with class-sensitive metrics.

The false positive rate is defined as

$$FPR = \frac{FP}{FP + TN} \quad (16)$$

This metric is central to the present study because false positives directly affect alert burden and triage usability. For multiclass evaluation, class-specific FPR values are calculated using a one-vs-rest formulation.

Matthews Correlation Coefficient is calculated as

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (17)$$

The methodological challenges of imbalanced learning are explicitly considered because skewed class distributions may inflate aggregate accuracy while concealing poor minority-class performance [53]. ROC-AUC is used to evaluate threshold-dependent class discrimination [54], while PR-AUC is reported where applicable because precision–recall evaluation is particularly informative when positive classes are rare or unevenly distributed [55]. MCC is included because it accounts for all confusion matrix components and remains informative under class imbalance [56]. For multiclass evaluation, the generalized multiclass MCC formulation implemented in the evaluation library is used.

False positive burden is analyzed separately from aggregate classification performance. This is necessary because two models may achieve similar F1-scores while producing substantially different alert volumes. Prior work on reducing false positives in network anomaly detection demonstrates that false alarm control is central to IDS usability [57]. Similarly, IDS alert classification research shows that detection outputs should be evaluated not only as predictions, but also as alert objects that require analyst handling [58]. The present study, therefore, reports class-specific false positive behavior and uses validation-based false positive rates as one component of the SFPS.

The experimental setup is implemented as a controlled public dataset benchmark. All models are trained and evaluated using the same preprocessed feature space, identical train–validation–test partitions, and the same evaluation protocol. This shared setup ensures that performance differences are attributable to model behavior and not to inconsistent preprocessing or data splitting.

All experiments are implemented in a controlled Python environment using standard machine learning and explainability libraries. The final manuscript reports the programming environment, library versions, hardware configuration, random seed control, preprocessing steps, split strategy, model hyperparameters, SHAP configuration, and SFPS weighting scheme. For Random Forest and XGBoost, the number of estimators, maximum depth, learning rate, regularization parameters, and class-weighting or imbalance-handling strategy are reported where applicable. For the lightweight deep learning baseline, the architecture, optimizer, learning rate, batch size, number of epochs, early stopping criteria, and parameter count are specified.

The study is interpreted as a controlled multiclass IDS and suspicious flow prioritization framework. It does not claim real-time SOC validation, full forensic reconstruction, or generalization across all IDS datasets. Its methodological contribution lies in integrating multiclass classification, SHAP-based explanation, validation-based false positive control, and suspicious flow prioritization under a transparent and reproducible evaluation design.

The implementation details summarized in Table 4 report the execution environment, preprocessing controls, model configuration, SHAP setup, and SFPS weighting scheme used for the final experiments. To assess the stability of the reported classification results, the final evaluation was repeated across five random seeds, and the results are reported as the mean $\pm$ standard deviation in the revised performance table.

Table 4. Reproducibility and implementation details.

Category	Specification Used in This Study
Programming environment	Python 3.12.10; Microsoft Windows 11 Pro.
Machine learning libraries	pandas 3.0.3; NumPy 2.4.5; scikit-learn 1.8.0; XGBoost 3.2.0.
Explainability library	SHAP 0.51.0.
Hardware configuration	Local workstation; Intel Core i7-8665U CPU (Intel Corporation, Santa Clara, CA, USA), 31.81 GB RAM; CPU-based execution.
Random seed control	Five random seeds were used for repeated-run evaluation: 42, 123, 2024, 2025, and 2026. The same preprocessing protocol, capped sampling logic, and stratified 70/15/15 split procedure were applied across runs.
Train/validation/test split	Stratified 70% training, 15% validation, and 15% testing split.
Normalization procedure	Z-score normalization was fitted on the training set and applied unchanged to the validation and test sets.
Models evaluated	Random Forest, XGBoost, and a compact, lightweight neural baseline.
Random Forest hyperparameters	n_estimators = 300; max_depth = None; class_weight = balanced_subsample; random_state varied according to the repeated-run seed; n_jobs = −1.
XGBoost hyperparameters	objective = multi:softprob; n_estimators = 300; max_depth = 6; learning_rate = 0.05; subsample = 0.8; colsample_bytree = 0.8; eval_metric = mlogloss; random_state varied according to the repeated-run seed.
Lightweight neural baseline architecture	Compact multilayer perceptron with two hidden layers, ReLU activation, dropout regularization, and softmax output layer for multiclass prediction.
Lightweight neural baseline training setup	Adam optimizer; batch size = 256; maximum epochs = 50; early stopping based on validation loss; trained as an unweighted compact baseline under the shared capped and stratified data protocol.

Table 4. Cont.

Category	Specification Used in This Study
SHAP configuration	Global SHAP analysis for feature-importance interpretation and local SHAP explanations for suspicious flow ranking; top contributing features retained for explanation strength computation.
SFPS weighting scheme	Weighted combination of model confidence, normalized SHAP explanation strength, attack severity weight, and validation-based false positive control adjustment.
Experimental dataset	Stratified capped experimental subset constructed from compatible CSE-CIC-IDS2018-derived sources; maximum 20,000 records per class, with rare classes retained in full.
Final retained records	213,463 records across one benign class and fourteen attack classes.

4. Results

The evaluation proceeds in five stages: repeated-run multiclass classification performance, minority-class and per-class performance, false positive behavior, SHAP-based explanation, and suspicious flow prioritization.

4.1. Multiclass Classification Performance

Table 5 reports the repeated-run multiclass classification performance of the three evaluated detection models on the retained experimental subset. Each model was evaluated across five random seeds, and the results are reported as mean $\pm$ standard deviation. This repeated-run design was used to assess whether the observed model differences remained stable under controlled variation in sampling, splitting, and model initialization.

Table 5. Repeated-run multiclass classification performance, mean $\pm$ SD across five random seeds.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	MCC	Macro ROC-AUC	Macro PR-AUC
Random Forest	0.9444 $\pm$ 0.0016	0.8359 $\pm$ 0.0070	0.8365 $\pm$ 0.0140	0.8355 $\pm$ 0.0104	0.9444 $\pm$ 0.0016	0.9372 $\pm$ 0.0018	0.9965 $\pm$ 0.0015	0.8794 $\pm$ 0.0084
XGBoost	0.9494 $\pm$ 0.0011	0.8443 $\pm$ 0.0136	0.8330 $\pm$ 0.0228	0.8366 $\pm$ 0.0193	0.9494 $\pm$ 0.0011	0.9429 $\pm$ 0.0012	0.9981 $\pm$ 0.0002	0.8875 $\pm$ 0.0075
Lightweight MLP	0.9219 $\pm$ 0.0020	0.8343 $\pm$ 0.0248	0.7937 $\pm$ 0.0130	0.7950 $\pm$ 0.0152	0.9211 $\pm$ 0.0022	0.9124 $\pm$ 0.0021	0.9949 $\pm$ 0.0015	0.8560 $\pm$ 0.0188

XGBoost achieved the strongest mean performance across most aggregate indicators, with an accuracy of 0.9494 ± 0.0011 , macro precision of 0.8443 ± 0.0136 , macro recall of 0.8330 ± 0.0228 , macro F1-score of 0.8366 ± 0.0193 , weighted F1-score of 0.9494 ± 0.0011 , MCC of 0.9429 ± 0.0012 , macro ROC-AUC of 0.9981 ± 0.0002 , and macro PR-AUC of 0.8875 ± 0.0075 . Random Forest produced closely comparable results, with an accuracy of 0.9444 ± 0.0016 , a macro F1-score of 0.8355 ± 0.0104 , a weighted F1-score of 0.9444 ± 0.0016 , and an MCC of 0.9372 ± 0.0018 . The lightweight MLP baseline remained lower on aggregate and macro-level indicators, with an accuracy of 0.9219 ± 0.0020 , macro F1-score of 0.7950 ± 0.0152 , and MCC of 0.9124 ± 0.0021 , confirming its role as a compact neural comparator rather than the strongest detection backbone in this framework.

These results support the use of tree-based ensemble models as the primary detection backbones for the subsequent explanation and prioritization stages. The comparison is not interpreted as a universal state-of-the-art ranking claim. Instead, it shows that both XGBoost and Random Forest provide stable multiclass detection outputs for examining the added analytical value of SHAP-style explanation and SFPS-based suspicious flow prioritization.

To complement the numerical comparison in Table 5, Figure 3 visualizes the repeated-run performance profile of the three evaluated models across all reported metrics. The graphical comparison shows that Random Forest and XGBoost produced closely aligned

overall performance, with XGBoost achieving the strongest mean aggregate profile across accuracy, weighted F1, MCC, macro ROC-AUC, and macro PR-AUC, while Random Forest produced the highest mean macro recall. The lightweight MLP baseline remained consistently lower on aggregate and macro-level indicators, which is consistent with its unweighted compact-baseline role under the shared capped and stratified data protocol.

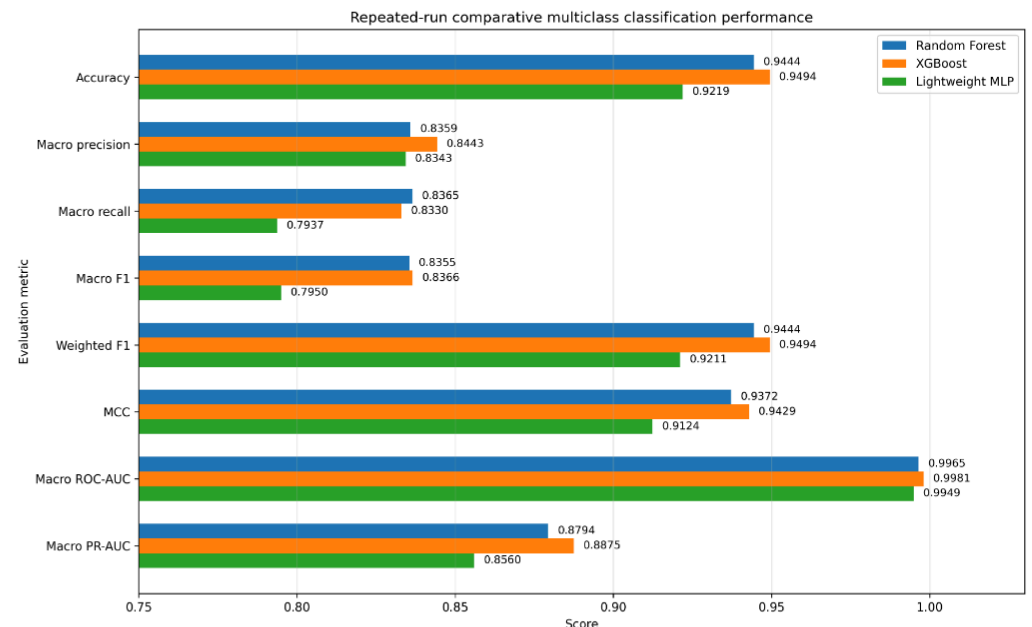


Figure 3. Repeated-run multiclass classification performance across evaluated models.

4.2. Minority-Class and Per-Class Performance

Because the retained dataset includes both capped high-frequency classes and substantially smaller minority classes, per-class behavior was examined separately from aggregate performance. Table 6 reports selected minority-class results for the final test split, focusing on classes whose interpretation is most sensitive to low support or attack-specific variability. This analysis is particularly important for SQL Injection, which contains only 87 retained records in the full experimental subset and 13 records in the final test split.

Table 6. Minority-class per-class performance on the final test split.

Class	Test Support	RF Precision	RF Recall	RF F1	XGBoost Precision	XGBoost Recall	XGBoost F1	MLP Precision	MLP Recall	MLP F1
Brute Force-Web	92	0.8485	0.9130	0.8796	0.9444	0.9239	0.9341	0.9180	0.6087	0.7320
Brute Force-XSS	34	1.0000	0.8824	0.9375	0.9688	0.9118	0.9394	1.0000	0.4706	0.6400
SQL Injection	13	0.6923	0.6923	0.6923	0.8571	0.4615	0.6000	0.6667	0.3077	0.4211

The results show that minority-class performance differed across models and attack categories. XGBoost achieved the strongest F1-score for Brute Force-Web and Brute Force-XSS, while Random Forest produced the strongest F1-score for SQL Injection. SQL Injection was detected with an F1-score of 0.6923 by Random Forest, 0.6000 by XGBoost, and 0.4211 by the lightweight MLP. These values indicate that the class was not ignored by the models, but they also show that reliability for this category remains limited by very low test-set support. Therefore, SQL Injection results are interpreted as minority-class evidence rather than as a basis for broad operational generalization.

The broader per-class results also show that high-frequency and structurally distinctive classes such as Bot, DDoS attacks-LOIC-HTTP, DoS attacks-Hulk, DDoS attack-HOIC,

and SSH-Bruteforce were classified with near-perfect F1-scores by the tree-based models. In contrast, minority or behaviorally overlapping categories required more cautious interpretation. This confirms the need to report macro-averaged metrics and per-class evidence alongside aggregate accuracy, particularly when evaluating IDS usefulness under class imbalance.

4.3. Class-Specific False Positive Burden

Although the three models achieved relatively strong aggregate classification performance, the false positive analysis shows that operational alert burden was not evenly distributed across classes. Table 7 reports the five highest class-specific false positive burdens for each model, calculated using a one-vs-rest formulation. In this one-vs-rest formulation, false positives for attack classes indicate flows incorrectly assigned to a given attack class, whereas the benign-class entry indicates false assignment to the benign label rather than a conventional false alert. This analysis is important because multiclass IDS models may achieve similar macro-F1 values while still generating different volumes and types of false alerts.

Table 7. Highest class-specific false positive burden by model.

Model	Class	FP	FPR	Precision	Recall
Random Forest	FTP-BruteForce	1409	0.0486	0.6475	0.8627
Random Forest	Benign	879	0.0303	0.7121	0.7320
Random Forest	Infiltration	803	0.0277	0.7253	0.7067
Random Forest	DoS attacks-SlowHTTPTest	413	0.0142	0.7940	0.5307
Random Forest	DDoS attack-LOIC-UDP	7	0.0002	0.9738	1.0000
XGBoost	FTP-BruteForce	1404	0.0484	0.6476	0.8600
XGBoost	Benign	1229	0.0423	0.6571	0.7929
XGBoost	Infiltration	616	0.0212	0.7422	0.5910
XGBoost	DoS attacks-SlowHTTPTest	421	0.0145	0.7914	0.5323
XGBoost	DDoS attack-LOIC-UDP	7	0.0002	0.9738	1.0000
Lightweight MLP	DoS attacks-SlowHTTPTest	1991	0.0686	0.5992	0.9920
Lightweight MLP	Benign	1933	0.0665	0.5642	0.8428
Lightweight MLP	Infiltration	482	0.0166	0.6892	0.3563
Lightweight MLP	Brute Force-Web	31	0.0010	0.6869	0.7391
Lightweight MLP	FTP-BruteForce	24	0.0008	0.9768	0.3373

For Random Forest, the largest false positive burden was associated with FTP-BruteForce, which produced 1409 false positives and a class-specific FPR of 0.0486. The Benign label was the second most affected one-vs-rest class, with 879 false positives and an FPR of 0.0303, followed by Infiltration with 803 false positives and an FPR of 0.0277. XGBoost showed a similar pattern, with FTP-BruteForce again producing the highest number of false positives (1404; FPR = 0.0484), followed by Benign (1229; FPR = 0.0423) and Infiltration (616; FPR = 0.0212). In contrast, the lightweight MLP baseline produced the highest false positive burden for DoS attacks-SlowHTTPTest (1991; FPR = 0.0686) and Benign traffic (1933; FPR = 0.0665), indicating weaker operational separation between normal and suspicious flows.

These results confirm that false positive behavior cannot be inferred from aggregate performance metrics alone. XGBoost achieved the strongest mean aggregate performance across most repeated-run indicators, while Random Forest produced closely comparable results. However, the false positive analysis shows that aggregate performance alone does not determine operational alert burden. Random Forest still produced a non-negligible false positive burden for FTP-BruteForce, Benign, and Infiltration, while XGBoost maintained strong probability discrimination and a false positive pattern concentrated in a small num-

ber of operationally relevant classes. The MLP baseline showed a more problematic false positive profile, particularly for Benign and SlowHTTPTest, which supports its use only as a lightweight comparator rather than as the primary backbone for the prioritization layer.

The false positive analysis directly motivates the triage-oriented design of the proposed SFPS. If suspicious flows were ranked only by model confidence, classes with higher false positive tendencies could receive excessive operational attention despite being comparatively noisy. Therefore, the prioritization layer incorporates validation-based false positive control together with model confidence, explanation strength, and attack severity. This design explicitly connects classification performance with operational triage: the goal is not only to detect attack classes, but to rank suspicious flows in a way that is more useful for analyst review.

To complement the tabular summary in Table 7, Figure 4 visualizes the highest class-specific false positive burdens across the three evaluated models. This graphical view makes it easier to compare the concentration and magnitude of false positive burden across operationally relevant classes. In particular, it highlights that the most substantial false positive counts were associated with FTP-BruteForce and Benign for the tree-based models, whereas the lightweight MLP produced especially high false positive burden for DoS attacks, SlowHTTPTest, and Benign traffic.

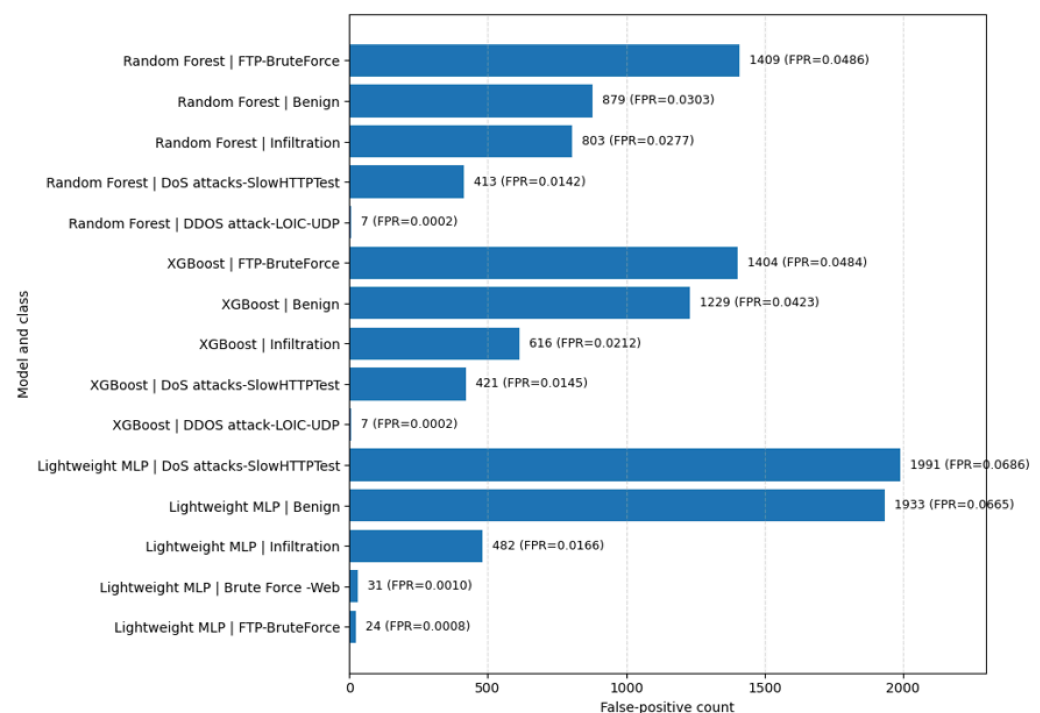


Figure 4. Highest class-specific false positive burden across evaluated models.

The figure visualizes the five highest class-specific false positive burdens for each evaluated model using a one-vs-rest formulation. The bar length represents the number of false positives, while labels report the corresponding class-specific false positive rate. The results show that false positive behavior was concentrated in a limited number of operationally relevant classes and differed across models, supporting the need for validation-based false positive control in the SFPS ranking layer.

The visualized false positive patterns reinforce the conclusion that aggregate classification performance alone is insufficient for assessing operational IDS usefulness. Although Random Forest and XGBoost achieved comparatively strong overall results, both still showed concentrated false positive burden in selected classes, while the lightweight MLP

exhibited a more problematic profile for normal and SlowHTTPTest traffic. These findings justify the inclusion of validation-based false positive control as one component of the proposed SFPS and provide a bridge to the SHAP-style explanation stage, where model outputs are examined as feature-level evidential structures.

4.4. SHAP-Based Feature Explanations

To examine how the classification model formed its multiclass predictions, an explanation layer was added after model training. Because the purpose of the proposed framework is not only to classify network flows, but also to support analyst-oriented triage, explanation was treated as an intermediate evidential layer between detection and prioritization. XGBoost native SHAP-style additive feature contributions were computed on a representative test-set sample of 1500 records. This sample size was selected to provide class-diverse explanation evidence while keeping the SHAP computation tractable on a CPU-based local workstation. XGBoost was selected for this explanation step because it achieved the strongest aggregate and probability-discrimination profile in the repeated-run classification results. To assess the stability of the explanation layer, the SHAP-style feature-ranking procedure was repeated across five independent 1500-record test-set subsamples, and the resulting top 15 feature rankings were compared.

The global explanation results are reported in Table 8. The values represent the mean absolute additive feature contributions averaged across the sampled test instances and multiclass output structure. Therefore, higher values indicate features that contributed more strongly, on average, to the model's class-discrimination behavior. These values should not be interpreted as causal effects or as standalone forensic evidence; rather, they indicate which flow-level characteristics most strongly shaped the model's predictions within the retained experimental subset.

Table 8. Global SHAP-style feature importance for XGBoost.

Rank	Feature	Mean Absolute Contribution
1	Dst Port	0.905653
2	Fwd Seg Size Min	0.666762
3	Init Fwd Win Byts	0.656421
4	Bwd Pkts/s	0.203707
5	Fwd IAT Min	0.178280
6	Flow IAT Mean	0.166628
7	Flow Pkts/s	0.148597
8	Init Bwd Win Byts	0.139687
9	Fwd Pkts/s	0.107508
10	Flow Duration	0.105413
11	Flow IAT Max	0.100373
12	Tot Fwd Pkts	0.096503
13	Flow IAT Min	0.096131
14	Tot Bwd Pkts	0.092454
15	TotLen Fwd Pkts	0.088225

To complement the ranked values in Table 8, Figure 5 visualizes the top 15 global XGBoost SHAP-style feature contributions. This graphical representation highlights the steep difference between the three highest-ranked features and the remaining flow-level indicators, suggesting that destination port, forward segment size, and initial forward window bytes were especially influential in the model's multiclass discrimination behavior.

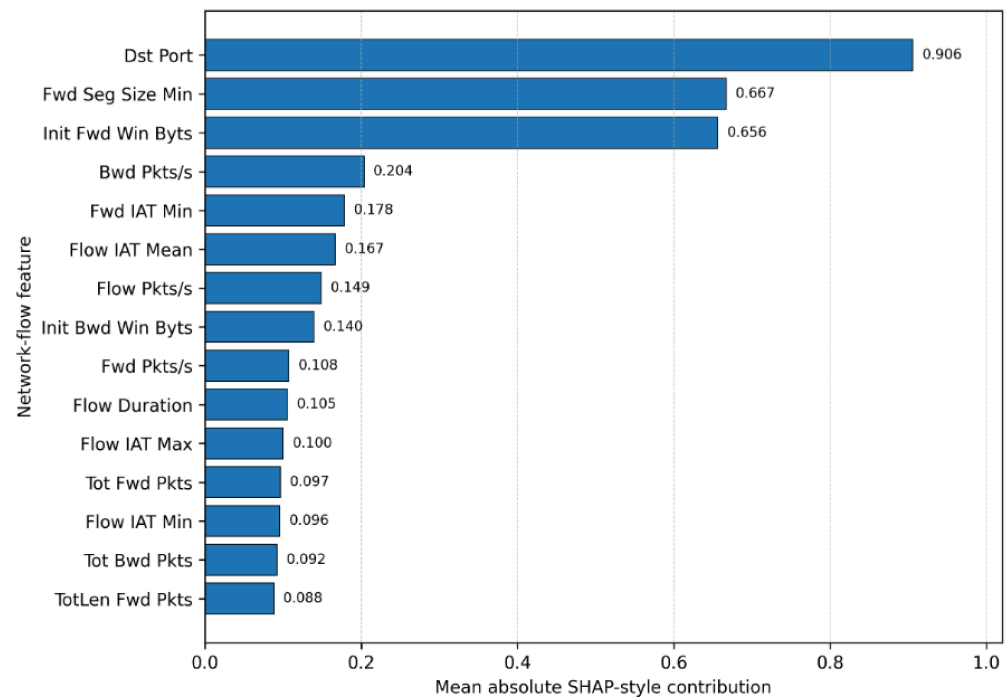


Figure 5. Global SHAP-style feature contribution ranking for the XGBoost model.

The figure shows the top 15 network-flow features ranked by mean absolute XGBoost SHAP-style additive contribution. Higher values indicate stronger average contribution to the model's multiclass discrimination behavior within the explanation sample. The values should be interpreted as feature-level evidential support for model predictions, not as causal effects or standalone forensic proof. To assess whether the top SHAP-style feature ranking depended on a single explanation sample, a stability check was conducted across five independent 1500-record test-set subsamples. Table 9 reports the mean rank, rank standard deviation, number of appearances in the top 15 list, and mean absolute contribution for the stable top features.

Table 9. SHAP-style feature-ranking stability across five independent 1500-record test-set subsamples.

Feature	Mean Rank	Rank SD	Top 15 Appearances	Mean Absolute Contribution
Fwd Seg Size Min	1.0	0.0	5/5	0.925885
Dst Port	2.0	0.0	5/5	0.858905
Init Fwd Win Byts	3.0	0.0	5/5	0.533462
Bwd Pkts/s	4.0	0.0	5/5	0.266677
Fwd Act Data Pkts	5.0	0.0	5/5	0.239779
Flow Duration	6.0	0.0	5/5	0.195491
Pkt Len Std	7.4	0.5477	5/5	0.188101
Flow IAT Mean	7.6	0.5477	5/5	0.188098
Bwd Pkt Len Std	9.0	0.0	5/5	0.179199
Fwd IAT Min	10.0	0.0	5/5	0.154429
Flow IAT Min	11.0	0.0	5/5	0.144777
Fwd Pkts/s	12.0	0.0	5/5	0.113123
Fwd Header Len	13.0	0.0	5/5	0.108008
Flow Pkts/s	14.0	0.0	5/5	0.100324
Flow IAT Max	15.0	0.0	5/5	0.093848

The stability check shows that all reported features appeared in the top 15 list across all five subsamples. The first six features retained identical ranks in all runs, and most remaining features also showed zero rank variability. Only Pkt Len Std and Flow IAT

Mean exchanged adjacent positions, with a small rank standard deviation of 0.5477. These results indicate that the explanation layer was not driven by a single 1500-record sample, but reflected a stable feature-ranking pattern across repeated explanation subsamples. The SHAP-style values are interpreted as model-specific evidential support for classification behavior, not as causal effects or standalone forensic proof. In the main explanation sample visualized in Figure 5, destination port had the highest mean absolute contribution, while the stability analysis in Table 9 shows that forward segment size and destination port consistently occupied the two highest-ranking positions across the repeated subsamples. Overall, the explanation results show that the model relied primarily on port-related, segment size, window size, packet rate, flow duration, inter-arrival time, and packet volume characteristics. This pattern is consistent with the structure of flow-based IDS data, where attack classes are often differentiated not by content inspection, but by statistical properties of connection behavior, packet timing, packet directionality, and traffic intensity.

The dominance of Dst Port, Fwd Seg Size Min, and Init Fwd Win Byts suggests that protocol-level communication structure and transport window behavior were central to the model's class separation. At the same time, the presence of Bwd Pkts/s, Flow Pkts/s, Fwd Pkts/s, and total packet count features indicates that traffic intensity and directional packet dynamics also contributed substantially to classification decisions. The inter-arrival time variables further show that temporal flow behavior was important for distinguishing between attack categories with different rhythm, duration, and packet spacing patterns. These findings provide the empirical basis for using explanation strength in the suspicious flow prioritization stage. In the proposed framework, SHAP-style contributions are not used merely as a post hoc interpretability add-on. Instead, their aggregated magnitude is used to estimate how strongly the model's prediction is supported by identifiable flow-level evidence. This is important because two flows may receive similar class probabilities while differing substantially in the clarity of the feature-level evidence supporting those predictions. The explanation layer therefore helps distinguish between high-confidence but weakly explained outputs and high-confidence outputs supported by stronger, more concentrated feature contributions.

Accordingly, the SHAP-based explanation step directly supports the SFPS design. Model confidence captures how strongly the classifier assigns a flow to a predicted class, while explanation strength captures how clearly the prediction is supported by feature-level contributions. Combined with attack severity and validation-based false positive control, this enables suspicious flows to be ranked not only by predicted probability, but by a broader triage logic that incorporates predictive confidence, interpretability, operational risk, and the expected false positive burden.

4.5. Suspicious Flow Priority Ranking

The final evaluation stage examined whether suspicious flow ordering changes when model confidence is combined with explanation strength, attack severity, and validation-based false positive control. This step directly addresses the operational motivation of the study: IDS outputs are not only classification decisions, but candidate alerts that must be ordered for analyst review. A probability-only ranking prioritizes flows solely according to the maximum predicted class probability. In contrast, the proposed Suspicious Flow Priority Score (SFPS) ranks suspicious flows using a composite triage logic that incorporates four components: model confidence, normalized explanation strength, attack severity weight, and validation-based false positive control.

For this analysis, the XGBoost model was used because it produced the strongest probability-discrimination profile in the classification results and was already used for the SHAP-style contribution analysis. Suspicious-flow candidates were defined as test-set

instances predicted as non-benign. The analysis was conducted on the top 5000 suspicious candidates ranked by model confidence. Native XGBoost additive feature contributions were used to estimate explanation strength for each candidate flow. The SFPS was then computed as a weighted combination of confidence, explanation strength, attack severity, and false positive control.

This comparison was designed to test whether SFPS changes the operational ordering of suspicious flows rather than whether it improves classification accuracy. Therefore, the relevant outcome is not a change in conventional detection metrics, but the degree to which the composite score reorders candidate alerts when explanatory support and validation-based reliability are added to model confidence.

To make this reordering effect explicit, Table 10 reports the overlap and Jaccard similarity between probability-only and SFPS-based rankings across top-k thresholds of 10, 25, 50, and 100. The results show that the two ranking strategies produced no shared flows at any evaluated threshold, with overlap values of 0/10, 0/25, 0/50, and 0/100 and Jaccard similarity equal to 0.0000 in all cases. This comparison is interpreted as evidence of post-detection ranking differentiation, not as proof that SFPS identifies objectively more urgent real-world incidents.

Table 10. Probability-only and SFPS-based ranking comparison with top-k overlap and Jaccard similarity.

Metric	Value
Number of suspicious candidates	5000
Top 10 overlap between probability-only and SFPS ranking	0/10
Top 10 Jaccard similarity	0.0000
Top 25 overlap between probability-only and SFPS ranking	0/25
Top 25 Jaccard similarity	0.0000
Top 50 overlap between probability-only and SFPS ranking	0/50
Top 50 Jaccard similarity	0.0000
Top 100 overlap between probability-only and SFPS ranking	0/100
Top 100 Jaccard similarity	0.0000
Mean probability-only confidence, top 15	0.999974
Mean SFPS, top 15	0.959845
Mean explanation strength, SFPS top 15	0.999523
Mean validation FPR, SFPS top 15	0.000000

Figure 6 visualizes this result as a ranking-set separation matrix. For each evaluated top-k threshold, the figure distinguishes between flows unique to the probability-only ranking, flows shared by both rankings, and flows unique to the SFPS-based ranking. The shared-flow component remains zero across all thresholds, while both ranking strategies retain distinct top-k sets. This confirms that SFPS did not reproduce the model confidence ordering but generated a different suspicious flow ranking by combining confidence, explanation strength, predefined severity, and validation-based false positive control.

The central empirical result of the prioritization stage is therefore ranking differentiation: adding explanation strength, severity weighting, and validation-based false positive control materially changed which suspicious flows would be presented first for analyst review. This result should not be interpreted as evidence that SFPS is objectively superior to probability ranking in real-world incident response, because the dataset does not contain SOC-confirmed priority labels. Instead, SFPS provides a transparent post-detection mechanism for moving from confidence-only ordering toward evidence- and reliability-aware suspicious flow prioritization.

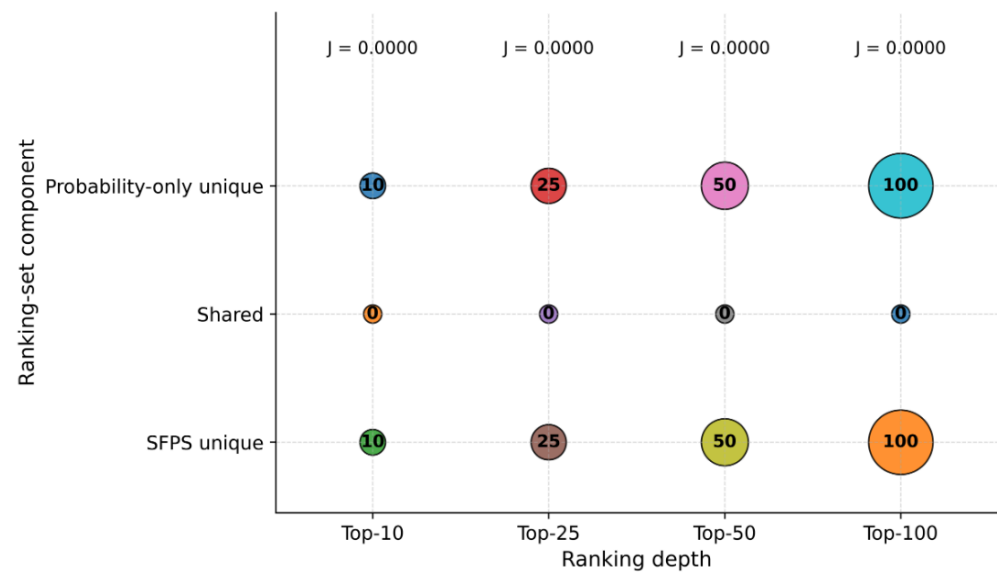


Figure 6. Bubble matrix visualization of ranking-set separation between probability-only and SFPS-based suspicious flow prioritization across top-k thresholds. Bubble size is proportional to the number of flows in each ranking-set component, while colors are used only for visual distinction and do not encode an additional quantitative variable.

To examine whether the observed reordering effect depended exclusively on the equal-weight SFPS configuration, an additional sensitivity check was conducted using four alternative weighting schemes. These schemes increased the relative contribution of model confidence, explanation strength, attack severity, or validation-based false positive control while keeping the remaining score components active. The purpose of this analysis was not to identify an optimal operational weighting strategy, but to assess whether SFPS-based prioritization remained distinct from probability-only ranking under plausible alternative configurations.

As shown in Table 11, the sensitivity analysis confirms that the SFPS reordering effect was stable across the tested weighting configurations. In all alternative schemes, the overlap with probability-only ranking remained 0/50 at the top 50 threshold, and the dominant top 50 class remained DoS attacks-Hulk. This indicates that the observed reordering was not merely an artifact of the equal-weight formulation. Rather, it reflected the broader effect of combining model confidence with explanation strength, attack severity, and validation-based false positive control. At the same time, the analysis should be interpreted as a robustness check rather than as deployment-level score calibration. Different operational environments may require different weighting schemes depending on analyst capacity, threat model, and acceptable false positive burden.

Table 11. SFPS sensitivity analysis across alternative weighting schemes.

SFPS Weighting Scheme	Weights	Top 50 Overlap with Probability-Only	Dominant Top 50 Class	Dominant Class Count	Mean Score, Top 50	Mean Confidence, Top 50	Mean Explanation Strength, Top 50	Mean Validation FPR, Top 50
Equal-weight	(0.25, 0.25, 0.25, 0.25)	0	DoS attacks-Hulk	50	0.949819	0.999966	0.999310	0.000000
Confidence-heavy	(0.40, 0.20, 0.20, 0.20)	0	DoS attacks-Hulk	50	0.959848	0.999966	0.999310	0.000000
Explanation-heavy	(0.20, 0.40, 0.20, 0.20)	0	DoS attacks-Hulk	50	0.959717	0.999966	0.999310	0.000000
Severity-heavy	(0.20, 0.20, 0.40, 0.20)	0	DoS attacks-Hulk	50	0.919855	0.999966	0.999310	0.000000
False positive-control-heavy	(0.20, 0.20, 0.20, 0.40)	0	DoS attacks-Hulk	50	0.959855	0.999966	0.999310	0.000000

The prioritization results show that multiclass IDS outputs can be transformed into a more triage-oriented suspicious flow ranking when prediction confidence is combined with explanation strength, attack severity, and validation-based false positive control. The purpose of SFPS is not to replace classification metrics, improve detection accuracy, or claim universal superiority over probability ranking. Its contribution is narrower and operationally specific: it provides a transparent post-detection mechanism for reordering suspicious flows according to confidence, evidential support, severity, and expected false positive reliability.

5. Discussion

The findings of this study should be interpreted as evidence from a controlled public dataset benchmark rather than as deployment validation or a universal model-ranking claim. The purpose of the proposed framework was not to introduce another high-complexity IDS architecture, but to examine whether multiclass IDS outputs can be extended into explainable, false positive-aware, and analyst-oriented suspicious flow rankings. In this respect, the results support the central premise of the study: detection performance alone is not sufficient for operational IDS usefulness when predicted flows must be interpreted, prioritized, and reviewed under alert-burden constraints.

The multiclass classification results showed that the two tree-based models provided a stable predictive basis for the subsequent explanation and prioritization stages. XGBoost achieved the strongest mean performance across most repeated-run aggregate indicators, including accuracy, macro F1-score, weighted F1-score, MCC, macro ROC-AUC, and macro PR-AUC, while Random Forest produced closely comparable results. This pattern supports the use of tree-based ensemble models as stable detection backbones for the subsequent explanation and prioritization stages. This pattern is consistent with the methodological role of tree-based ensemble models in structured-data classification, where nonlinear feature interactions and robust aggregation can support stable predictive behavior [46]. The relevance of boosted ensemble learning is also reflected in the XGBoost results, since gradient boosting and its scalable regularized implementation are well suited to tabular, flow-based classification settings in which probability estimates are required for downstream prioritization [47,48]. At the same time, these findings do not imply that Random Forest or XGBoost is universally superior to more complex IDS architectures. Rather, they show that computationally tractable models can provide sufficiently strong detection outputs for evaluating post-detection explanation and prioritization.

The lightweight MLP baseline produced lower macro-level performance than the two tree-based models, particularly in relation to macro recall and macro F1-score. This does not invalidate the relevance of lightweight or deep learning approaches in IDS research. Recent studies continue to show that recurrent, optimized, explainable, IoT-oriented, and multiclass deep learning frameworks can achieve strong results under appropriate data and model assumptions [49–52]. However, in the present benchmark, the compact neural baseline was used only as a controlled comparator, not as the main methodological contribution. Its lower performance reinforces the importance of matching model complexity and architecture to the structure of the flow-based dataset and to the downstream purpose of the study.

The interpretation of the classification metrics also needs to account for class imbalance and metric sensitivity. Because CSE-CIC-IDS2018 contains uneven class distributions, accuracy alone is insufficient for evaluating multiclass IDS behavior. The use of macro F1-score, weighted F1-score, MCC, ROC-AUC, and PR-AUC provides a more balanced view of model performance under imbalanced conditions, especially when rare attack classes and uneven positive-class distributions affect aggregate results [53–56]. In this study, XGBoost

provided the strongest mean aggregate profile across most repeated-run indicators, while Random Forest remained closely comparable and retained strong performance on selected minority-class results. This distinction is important because XGBoost was subsequently used not only as a classifier, but also as the basis for explanation and suspicious flow prioritization. The false positive analysis is particularly important for interpreting the operational meaning of the classification results. Although aggregate performance metrics were relatively strong, false positive burden was concentrated in selected classes and differed across models. This confirms that multiclass IDS evaluation cannot rely solely on accuracy, macro-F1, or area-under-curve indicators. A model may perform well in aggregate while still generating class-specific false positive patterns that affect analyst workload and alert usability. Prior work on reducing false positives in network anomaly detection and on IDS alert classification similarly shows that false alarms should be treated as operational objects requiring explicit analysis, not merely as residual classification errors [57,58]. In the present study, this issue is addressed by incorporating validation-based false positive control directly into the SFPS ranking layer.

The SHAP-style explanation results further show that the model's class-discrimination behavior was supported by interpretable flow-level evidence. The strongest global contributors included destination port, forward segment-size minimum, initial forward window bytes, packet rate variables, flow duration indicators, and inter-arrival time features. These results are consistent with the flow-based structure of CSE-CIC-IDS2018, where attack categories are distinguished through statistical traffic behavior rather than content inspection. More importantly, the explanation layer was not treated merely as a post hoc visualization tool. Instead, explanation strength was used as one component of the suspicious flow ranking mechanism. This extends the role of XAI in IDS from descriptive interpretation toward evidence-aware prioritization, while remaining consistent with the broader literature that positions explainability as necessary for transparency, trust, and analyst-facing cybersecurity decision support [30,32–37].

5.1. SFPS as a Post-Detection Prioritization Mechanism

The main contribution of the proposed framework is the transformation of multiclass IDS outputs into a triage-oriented ranking rather than a new classification metric. The SFPS combines four complementary dimensions: model confidence, explanation strength, attack severity, and validation-based false positive control. This design reflects the operational distinction between detecting a suspicious flow and deciding which suspicious flows should be reviewed first. In practical alert-handling contexts, high model confidence alone may be insufficient if the predicted class is associated with higher false positive tendencies or if the prediction is weakly supported by identifiable feature-level evidence.

The prioritization results demonstrate this distinction clearly. Probability-only ranking and SFPS-based ranking produced no shared flows at the evaluated top-k thresholds, including $k = 10, 25, 50$, and 100 . Probability-only prioritization selected SSH-Bruteforce flows in the top 50, whereas SFPS prioritized DoS attacks-Hulk flows. This shift reflects the combined influence of near-maximal explanation strength, higher predefined attack severity, and zero validation-set false positive rate in the composite score. The result should not be interpreted as evidence that SFPS improves detection accuracy or proves superior real-world incident prioritization. Instead, it shows that adding explanation strength and false positive reliability can materially change the order in which suspicious flows are presented for analyst review.

This finding is relevant to the detection-to-triage gap identified in prior work on security analysts, alert fatigue, alert prioritization, and intelligent alert management [6–9,41–45]. In conventional IDS evaluation, model outputs are often treated as final classification de-

cisions. In operational settings, however, they function as candidate alerts that must be filtered, contextualized, and prioritized. The SFPS addresses this intermediate layer by providing a transparent post-detection heuristic for ranking suspicious flows according to multiple evidence dimensions. It does not replace analyst judgment, alert correlation, or incident response workflows, but it offers a reproducible mechanism for converting multiclass detection outputs into a more reviewable order.

The top-k reordering result is therefore important because it provides direct empirical evidence that probability and triage priority are not equivalent within the tested benchmark setting. Two flows may both receive very high predicted probabilities, but they may differ in explanation strength, class severity, and validation-based false positive reliability. The proposed SFPS operationalizes this distinction by favoring flows that are not only confidently classified but also strongly supported by feature-level evidence and associated with lower expected false positive burden. This is especially relevant in alert-heavy environments, where analysts may not be able to inspect all model-generated detections with equal urgency. The result demonstrates a change in review ordering, not validated superiority in live SOC decision-making.

The framework also contributes to the role of explainability in IDS. Many XAI applications in cybersecurity focus on global feature importance or local explanation examples. While such outputs are useful, they often remain separate from alert-handling logic. In the present framework, SHAP-style contributions are integrated into the prioritization mechanism through explanation strength. This does not make the resulting score a measure of causal certainty or forensic proof. Rather, it provides a controlled way to distinguish predictions that are merely high-confidence from predictions that are high-confidence and supported by stronger feature-level evidence. This interpretation is consistent with the broader view of explainability as support for human oversight and decision-making, especially when automated systems affect operational trust and response prioritization.

At the same time, the SFPS should be understood as a scope-controlled ranking heuristic rather than as a universal risk score. Its components were deliberately simple and transparent: confidence, explanation strength, severity weight, and false positive adjustment. Equal weights were used to avoid tailoring the score to a specific operational environment. The additional sensitivity check showed that the top 50 reordering pattern remained stable under confidence-heavy, explanation-heavy, severity-heavy, and false positive-control-heavy configurations. This supports the robustness of the observed prioritization effect, but it does not remove the need for environment-specific calibration: different SOCs, sectors, or threat models may reasonably assign different importance to severity, false positive reliability, or explanation strength. The present results therefore demonstrate the feasibility and empirical effect of post-detection prioritization, not the optimal weighting strategy for all deployment contexts.

5.2. Limitations and Future Research

Several limitations should be considered when interpreting the findings. First, the study was conducted on a CSE-CIC-IDS2018-derived experimental subset rather than on live operational traffic. Although CSE-CIC-IDS2018 is a widely used public benchmark with diverse attack categories and flow-level features, benchmark datasets cannot fully reproduce the variability, adversarial adaptation, infrastructure dependencies, and organizational constraints of real security operations. The results should therefore be understood as controlled benchmark evidence rather than proof of deployment readiness. This limitation is consistent with broader concerns in IDS research regarding dataset representativeness, benchmark heterogeneity, and the effect of dataset structure on reported model performance [12,13,16–19].

Second, the study focused on a limited set of detection backbones: Random Forest, XGBoost, and a compact, lightweight MLP baseline. This choice was intentional because the aim was to evaluate a transparent detection-to-triage pipeline rather than to establish a broad cross-family leaderboard. More complex architectures, including graph-based, transformer-based, federated, or LLM-enhanced IDSs, were not experimentally benchmarked in this study. Such models may provide stronger representational capacity under different assumptions, but they also introduce additional complexity, computational cost, and evaluation requirements. The present framework should therefore be interpreted as complementary to, rather than competitive with, advanced IDS architectures such as dynamic graph or semantic-enhanced models [5].

Third, the SFPS uses predefined and equally weighted components. Equal weighting was selected to maintain transparency and avoid tuning the prioritization score to a specific operational scenario. However, in real SOC environments, the relative importance of model confidence, explanation strength, attack severity, and false positive reliability may differ by organization, sector, threat model, and analyst workflow. Future research should therefore examine alternative weighting schemes, sensitivity analysis, and validation-guided calibration of SFPS components. Such work could determine whether different environments require confidence-dominant, severity-dominant, explanation-dominant, or false positive-aware weighting configurations.

Fourth, the explanation layer should be interpreted as evidential rather than causal. SHAP-style additive contributions identify which flow-level features supported the model's prediction, but they do not prove causal attack mechanisms, establish forensic attribution, or replace incident investigation. This is particularly important because flow-based IDS datasets contain statistical traffic summaries rather than packet payloads, host-level telemetry, or complete attack-chain context. Future work should therefore combine feature-level explanations with richer forms of evidence, including temporal alert correlation, host-based context, threat intelligence, and analyst feedback.

Fifth, the prioritization experiment evaluated reordering behavior within a controlled test-set setting, not analyst performance in a live SOC. The observed difference between probability-only and SFPS-based ranking demonstrates that SFPS materially changed the post-detection ordering of suspicious flows across the evaluated top-k thresholds. The study does not measure whether the SFPS-ranked flows represent more urgent real-world incidents, nor whether analysts would resolve incidents faster, reduce cognitive load, or improve response quality when using SFPS-ranked alerts. Future research should therefore evaluate the framework in human-in-the-loop experiments, simulated SOC environments, or retrospective alert-review studies with confirmed incident-priority labels.

Sixth, the present evaluation was conducted on a static benchmark subset and does not test whether the same false positive-control behavior would remain stable under streaming traffic, concept drift, changing attack prevalence, or evolving operational baselines. Future work should therefore evaluate SFPS under streaming or near-real-time traffic conditions with temporal validation and drift-aware recalibration.

Future work could also examine whether SFPS-based prioritization can be aligned with operationally oriented threat-evaluation frameworks such as MITRE ATT&CK Evaluations [59]. Such integration would help connect suspicious flow ranking with adversary behavior categories, tactic–technique mappings, and more realistic evaluation scenarios. More broadly, future research should test the proposed framework across additional IDS datasets, model families, attack distributions, and deployment assumptions, consistent with recent reviews highlighting open challenges and future directions in machine-learning-based intrusion detection [60]. These extensions would allow the detection-to-triage framework to be evaluated beyond a single benchmark setting while preserving its central

premise: IDS outputs should not only detect attacks, but also support explainable, false positive-aware, and analyst-oriented prioritization. Future research could also examine how SFPS-based triage interacts with lightweight or cost-aware IDS architectures, including models designed around performance–risk trade-offs and false positive control in benchmark intrusion detection settings [61].

6. Conclusions

This study examined how multiclass intrusion detection outputs can be transformed from isolated classification decisions into explainable and triage-oriented suspicious flow rankings. Rather than proposing another high-complexity IDS architecture or pursuing a general state-of-the-art comparison, the study focused on the operational gap between detection and analyst review. Its central premise was that IDS outputs become more useful when they are not only classified, but also explained, evaluated in relation to false positive burden, and ranked according to review priority.

The results support this premise. Random Forest and XGBoost provided strong and stable multiclass detection backbones under the shared experimental protocol, while the lightweight MLP baseline showed a weaker macro-level profile. However, the findings also demonstrate that aggregate classification performance alone is insufficient for operational interpretation. False positive burden was concentrated in selected classes and differed across models, confirming that similar global performance can still produce different alert-handling implications. This reinforces the need to evaluate IDS outputs not only as predictions, but also as candidate alerts that may impose different levels of analyst burden.

A central contribution of this work lies in integrating explanation and false positive control into the post-detection stage. The SHAP-style explanation layer identified flow-level features that supported XGBoost multiclass discrimination, including destination port, forward segment-size minimum, initial forward window bytes, packet rate variables, and inter-arrival time indicators. These explanations were not treated merely as descriptive visualizations. Instead, their aggregated strength was used as one component of the Suspicious Flow Priority Score, linking model output to interpretable evidential support.

The SFPS results show that classification confidence and triage priority are not equivalent. Probability-only ranking and SFPS-based ranking produced different suspicious flow orderings across the evaluated top-k thresholds, indicating that the proposed score materially changed the order in which flows would be presented for review. This does not mean that SFPS improves detection accuracy, proves real-world incident urgency, or establishes deployment readiness. Its contribution is more specific: it provides a transparent post-detection mechanism for ranking suspicious flows by combining model confidence, explanation strength, attack severity, and validation-based false positive control. Future extensions could also examine whether stronger hybrid or IoT-oriented anomaly detection backbones can be integrated into the same post-detection triage logic, particularly as recent work has shown the relevance of CNN–BiGRU, optimization-based, and synthetic-balancing strategies for anomaly detection in IoT security environments [62].

In conclusion, this study contributes a scope-controlled detection-to-triage framework for multiclass IDS evaluation. It shows that public benchmark IDS outputs can be extended beyond conventional classification metrics into a more operationally meaningful evidence structure. The proposed framework does not replace analyst judgment, alert correlation, or incident response procedures. Instead, it offers a reproducible way to connect multiclass prediction, explanation, false positive reliability, and suspicious flow prioritization. These insights support a shift in IDS research from asking only which model detects best toward asking how detection outputs can be made explainable, reliable, and usable for analyst-oriented triage.

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Data Availability Statement: The CSE-CIC-IDS2018 dataset used in this study is publicly available from the Canadian Institute for Cybersecurity at <https://www.unb.ca/cic/datasets/ids-2018.html> (accessed on 16 May 2026). The experimental subset was constructed from compatible publicly available CSE-CIC-IDS2018-derived sources was described in Section 3. No new data were generated during the study. Additional implementation details can be provided by the corresponding author upon reasonable request.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

IDS	Intrusion Detection System
MCC	Matthews Correlation Coefficient
MLP	Multilayer Perceptron
PR-AUC	Area Under the Precision–Recall Curve
RF	Random Forest
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SFPS	Suspicious Flow Priority Score
SHAP	SHapley Additive exPlanations
SOC	Security Operations Center
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

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